The effect of property taxes on house prices: Evidence from border discontinuities in Chile

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

An overview

- **Question**: How do property taxes affect home prices?
- **Goal**: Quantify the effect using **Chilean** data.
- **Approach**: Exploit border differentials in property taxes to identify the effect through a **Difference-in-Discontinuities** (Diff-in-Disc) strategy.

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Motivation: Property tax capitalization

- You want to purchase a home. Two houses seem almost identical, but one of them is slightly less expensive, as it enjoys an annual tax burden that is 100,000 CLP lower.
- How much more would you be willing to pay for the low-tax house?
- **This question sits at the heart of standard capitalization theory and tax incidence.**

Literature: Theory of property tax capitalization

- The theory of local public finance argues that, as long as property taxes are capitalized to some extent, the effect of property taxes on home values should be strictly negative (Oates, 1969).
- Under full capitalization, a property that has a finite life span of n years will have a market value P_t in year t $(1 \leq t \leq n)$ given by

$$P_t = \sum_{s=t}^{n} \frac{Y_s - \tau P_s}{(1+i)^{s-t}}$$

  where Y_s is the inflow of property value in year s, i is the interest rate and τ the property tax rate.
- We can see that P_t decreases with τ: a property tax should have a negative effect on prices.

Literature: Empirics of property tax capitalization

- Testing the influence of property taxes on home prices involves several difficulties (Bai et al., 2014):
  - Causality can run from property values (P) to property taxes (τ).
    - ★ For instance, if the local government aims for a fixed tax revenue, it may impose lower tax rates on communities with higher home values.
  - Factors like local income (Y_t), inflation, public expectations, and quality of public services are challenging to control for.
    - ★ Y_t is linked to the quality of local public services, monetary policies, inflation, and public expectations (Poterba, 1984).
    - ★ Higher property tax rates, commonly used to finance local public services like in the US, are associated with higher property values (P).
    - ★ This association arises from the improvement in the quality of public goods (Rosen and Fullerton, 1977).

Literature: Empirics of property tax capitalization

- To mitigate biases from endogeneity issues, researchers usually resort to natural experiments resulting from exogenous policy changes.
- However, even with exogenous changes in property taxes (τ), it remains challenging to fully control for factors like Y_t, i, and others.
- According to empirical studies, there is no consensus on the degree of tax capitalization, and some studies even find a positive effect of property taxes on prices (Bai et al., 2014).

Contribution: Empirics of property tax capitalization

- Bai et al. (2014) studies two cities in China, finding that a property-tax experiment lowered the Shanghai average home price by 11%–15% but raised the Chongqing average home prices by 10%–12%.
  - They argue that the price increase in Chongqing can be driven by a spillover effect from high-end to low-end properties.
- Elinder and Persson (2017) show house prices did not respond to a substantial cut in the national property tax in Sweden. However, effects are heterogeneous, as they report price increases only in a small segment of the market containing properties with very high tax values.
- Oliviero and Scognamiglio (2019) show that a one standard deviation increase in municipal-level property tax intensity leads to a 2.7% reduction of property values in the year of an Italian tax reform.
- See Hilber (2017) for a review of the literature up to 2017.
- **Contribution: To test the impact of spatial tax differentials on home values.**

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Institutional setting

- Property taxation is based on the **fiscal valuation** of a property, which is a monetary value assigned by the SII (IRS) to all properties in the country.
- Every four years, the SII is responsible for carrying out a process called reassessment, where a new appraisal is conducted with the aim of updating the fiscal assessment of properties.
- The fiscal valuation A_f of a home is divided into two components:

$$A_f = A_{const} + A_{land} \tag{1}$$

  where A_{const} is the valuation of the construction and A_{land} is the valuation of the land.
- The construction valuation will depend on the square meters built, as well as the materiality of the construction and its associated quality level.

Institutional setting

- With respect to the value of the property's land, it consists of the square meters of the land and the value per square meter associated with the property's location.
- This value is obtained based on the homogeneous area (HA) in which the property is located, and according to the SII, it is calculated as follows:

$$A_{land} = TA_a \tag{2}$$

where T corresponds to the size of the property in square meters and A_a corresponds to the unit value of HA a.

Institutional setting

- According to the SII, homogeneous areas are subdivisions of communes aimed at grouping territorial spaces that share common urban characteristics.
- Based on these characteristics, the SII assigns a unit value to the homogeneous area. This means that this value does not depend on the property's characteristics but rather on its surroundings.
- The first appearance of homogeneous areas was in 2013. The methodology used to determine the unit value of these homogeneous areas is not public, nor is the justification for how the border limits are determined.
- **Homogeneous areas provides us with the opportunity to exploit tax differentials which are based on spatial location.**

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Data sources

We combine a set of diverse datasets/data sources:

1. **Cartography of homogeneous areas**
   - Retrieved from Correa (2018), who enabled through **Red CEDEUS** the corresponding cartography of the homogeneous areas for the **year 2015**.
   - Quilicura, Pedro Aguirre Cerda, El Bosque and Lampa municipalities are not included.
   - This document corresponds to a shapefile type file, which delineates the boundaries of the homogeneous areas (HA).

Data sources and assembly

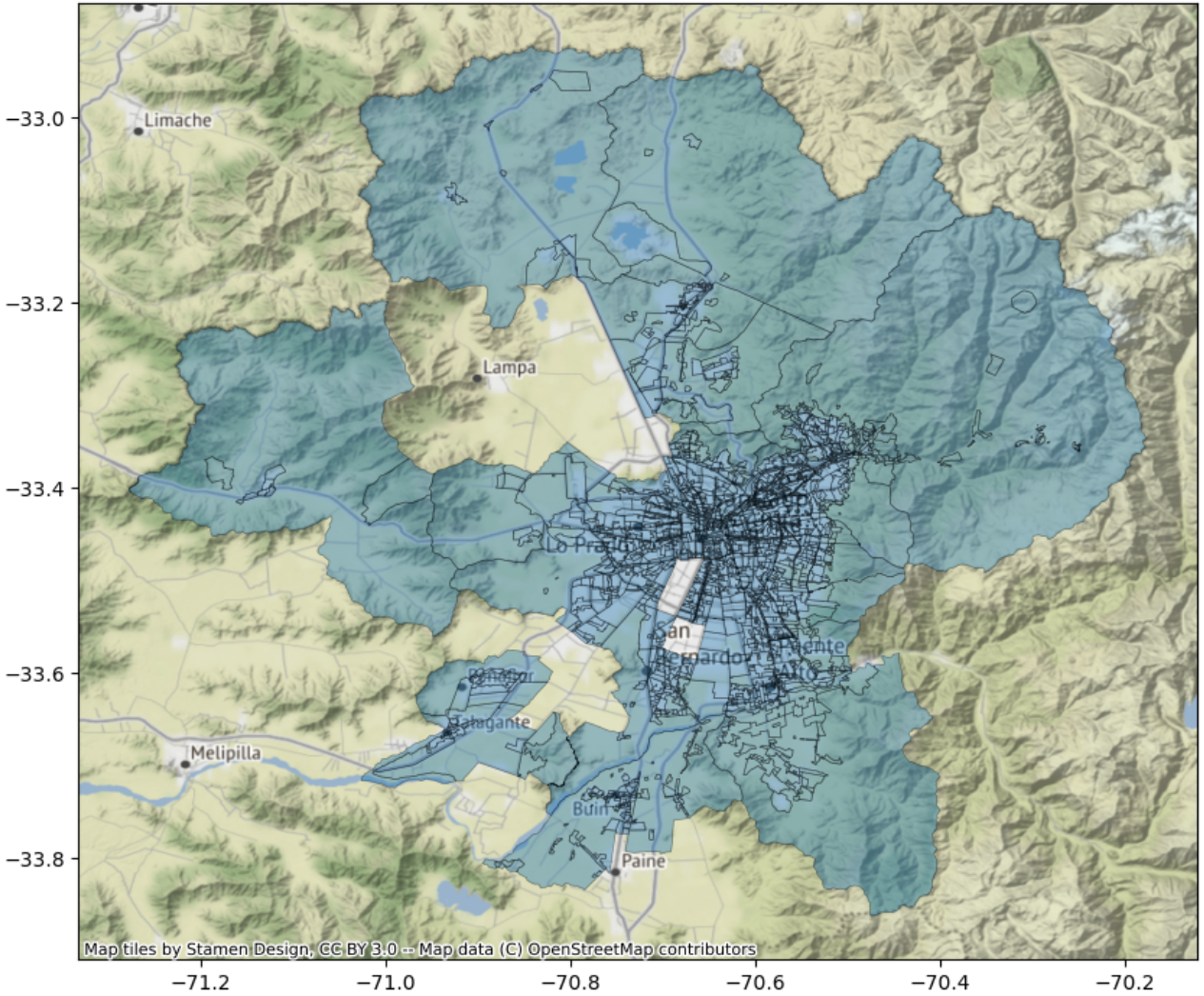

Figure 1: Cartography representing shapefile total coverage

Data sources and assembly

2. **Characteristics of real estate**
   - We retrieve this data from the **SII**.
   - Due to the *Ley de Transparencia*, The SII (Internal Revenue Service) is obliged to share data characterizing all property roles in Chile individually.
   - These data are used by the SII to calculate the fiscal appraisal of the property; therefore, any individual has the right to request them from the SII.
   - Among these data are the **property role, the commune, the fiscal appraisal, square meters built accompanied by its materiality and quality, square meters of land, and the type of destination of the role.**
   - For the purposes of the study, we will only focus on roles whose destination corresponds to Residential (H).

Data sources and assembly

3. **Coordinates of the real estate properties and appraisals in homogeneous areas**
   - We need property's coordinates to calculate the Euclidean distances to the borders of the adjacent homogeneous areas (AH).
   - We retrieve these coordinates through the **Cartografía Digital SII Mapas**, whose function is to provide public information about the fiscal appraisals of all properties in the country.
   - By selecting a role, the platform provides information about the address, fiscal appraisal, homogeneous area, real estate destination, and a map indicating the geographical coordinates.
   - Due to the enormous amount of data required to be collected, it was necessary to program an automated procedure that accesses the web platform, retrieves the data for a specific role, and saves it in a database.
   - This technique is called **web scraping** and is widely used for data collection from internet sources.

Data sources and assembly

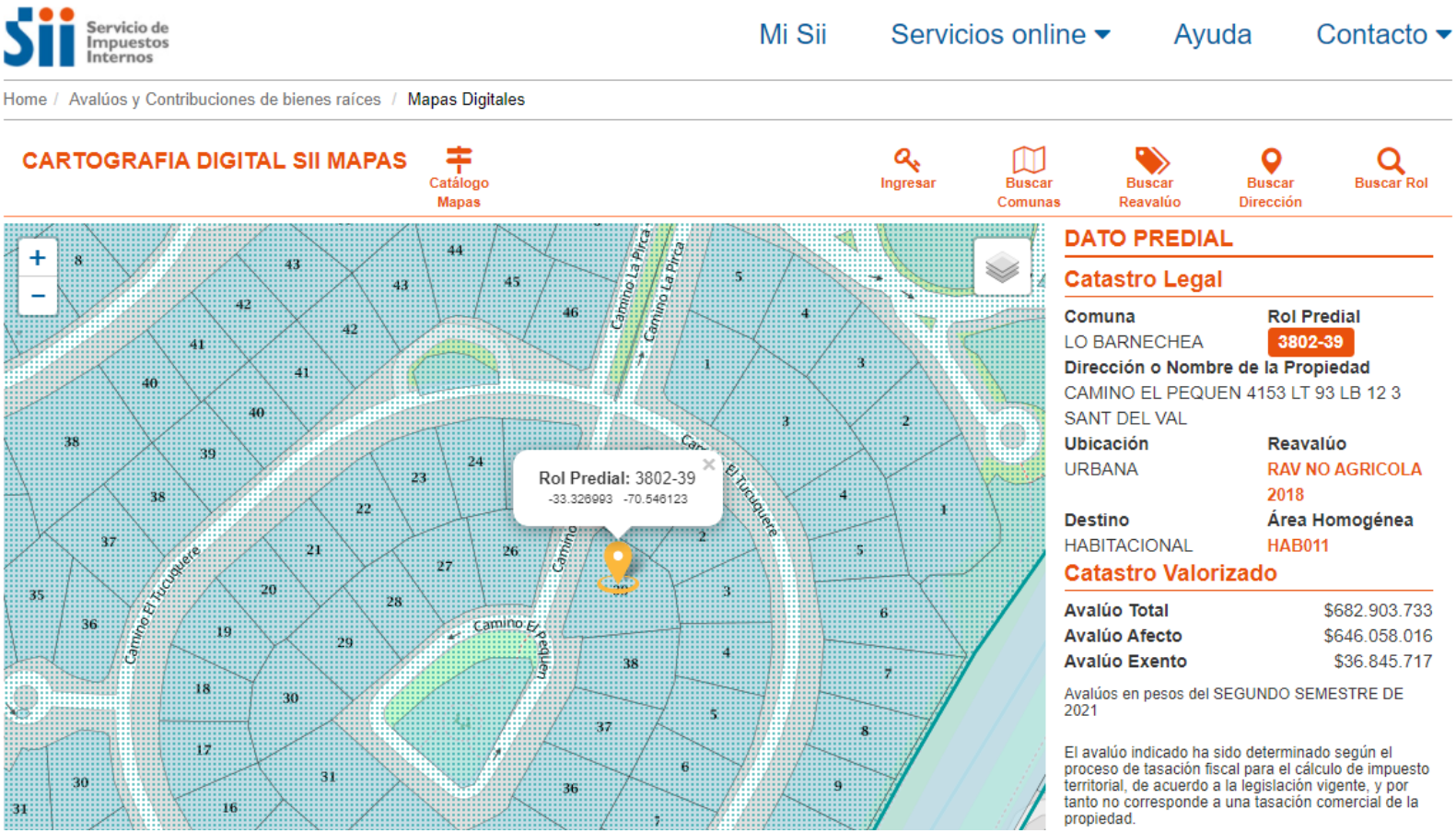

Figure 2: SII Digital Cartography Maps Tool

Data sources and assembly

3. **Coordinates of the real estate properties and appraisals in homogeneous areas**
   - The web scraping program was executed for each role contained in this list, and the results were stored in a database on AWS.
   - After 11 days, the procedure concluded, and a total of 1,090,282 geographical coordinates were collected for Greater Santiago.
   - The next Figure shows a random sample of 30,000 properties that were georeferenced using the data extraction procedure.

Data sources and assembly

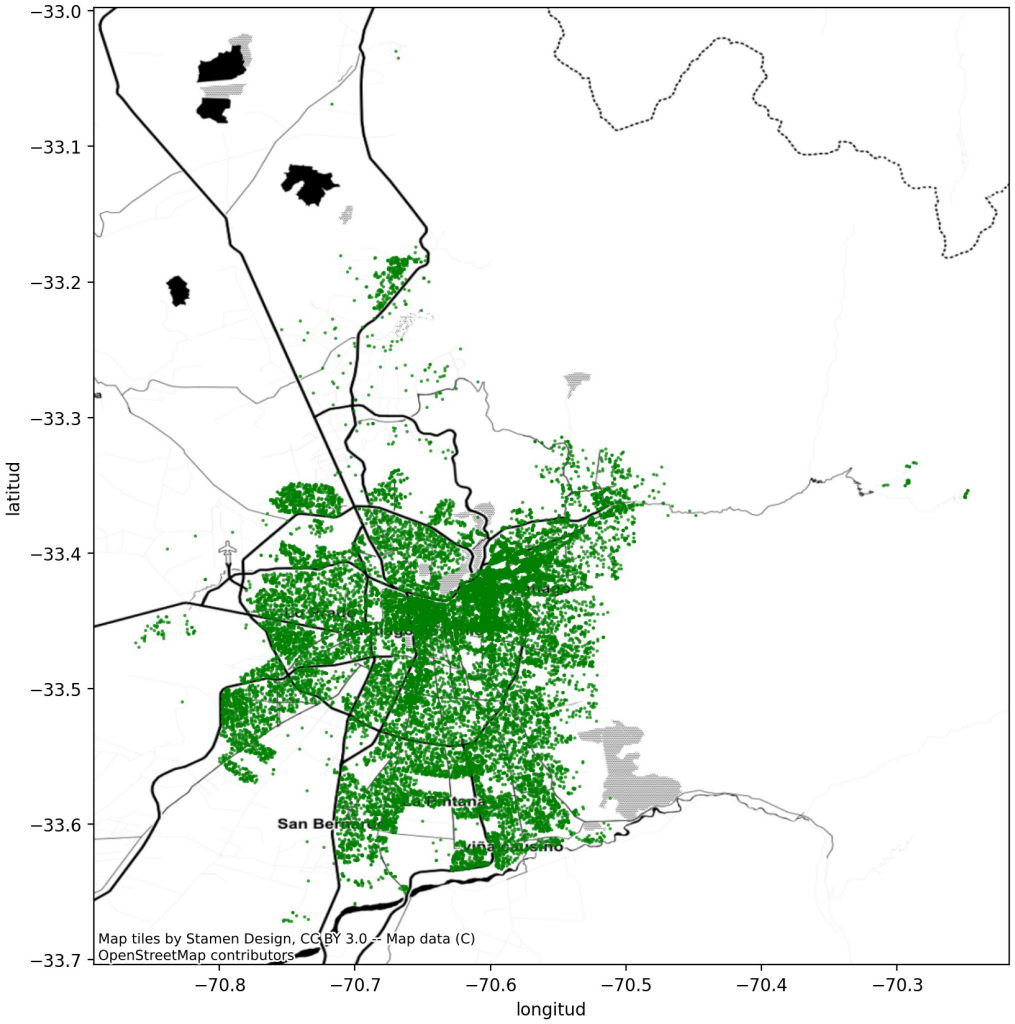

Figure 3: Random sample of 30,000 georeferenced properties

Data sources and assembly

3. **Market valuation of real estate properties**
   - The real estate conservators are the entities responsible for keeping records of all property market transactions in the country.
   - Within this registry, the value at which the property was traded is included. We refer to this as the property's market value.
   - **The Real Estate Studies Center at the ESE Business School of the Universidad de Los Andes** has a private database containing the value in UF (Unidad de Fomento, a monetary unit used in Chile) of purchase-and-sell agreements from the Conservador de Bienes Raíces de Santiago (CBRS).
   - We retrieve house prices from thus database, which contains records from 26 communes in the city of Santiago.

Definition of adjacent HAs

- We define adjacent HAs as those that **share a vertex**.

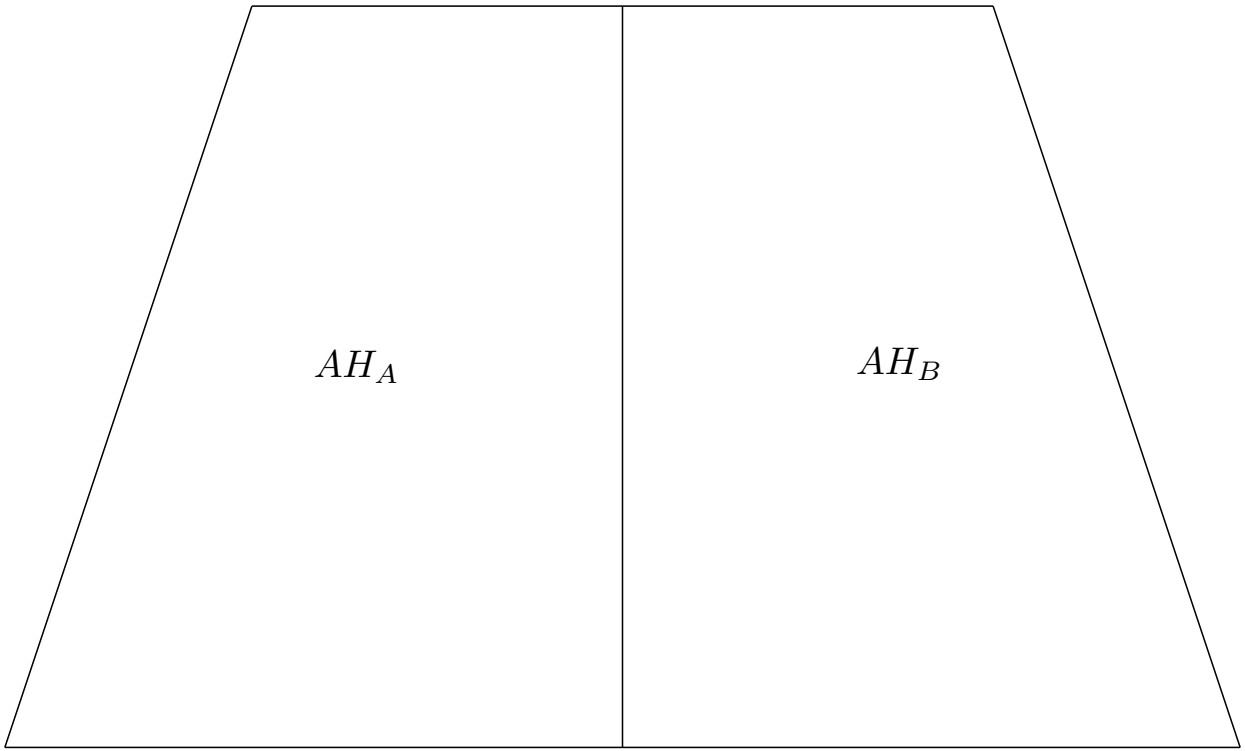

Figure 4: Stylized representation of adjacent homogeneous areas

Computation of distance to border

- Given that a HA may have more than one neighboring HA, it is possible for the same role to be assigned more than one distance, one for each neighboring HA.
- Once the distance is determined, it is assigned a positive (negative) polarity if the neighboring HA corresponds to one with a lower (higher) unit value.
- In regression discontinuity analyses, this polarity help identify on which side of the cutoff the observation lies.

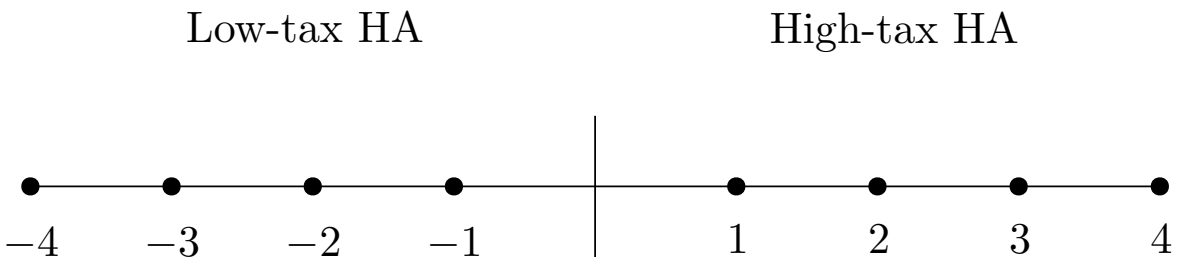

Figure 5: Polarity of distance to border

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Empirical strategy: Diff-in-Disc intuition

- We rely on a Difference-in-Discontinuities strategy. The identifying assumptions for this strategy were first formalized by Grembi et al. (2016). Later, Butts (2021) identified the necessary assumptions for the geographic Diff-in-Disc setting.

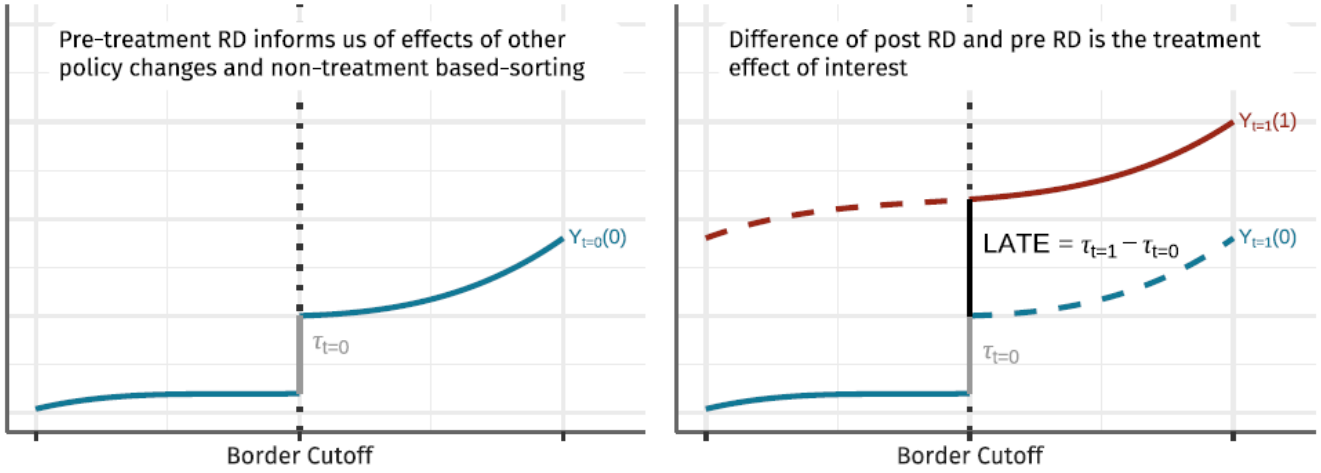

Figure 6: Stylized diff-in-disc identification, taken from Butts (2021).

- Under this framework, the RD estimated previous to the establishment of HAs is a measure of non-treatment-based sorting such as geographic invariant sorting and administrative sorting.

Empirical strategy: Identifying assumptions

- Potential outcomes for individual $i = 1, \ldots, n$ at time $t = 0, 1$ are assumed to follow

$$y_{it} = f_t(D_i) + \gamma(D_i)\mathbb{1}(D_i \geq 0) + \tau(D_i)\mathbb{1}(D_i \geq 0)\mathbb{1}(t = 1) + \epsilon_{it}, \quad (3)$$

- y_{it} is the dependent variable, log-prices, in our case.
- D_i is the running variable, distance from a home to an adjacent homogeneous area.
- $f_t(D_i)$ is an untreated location-specific component.
- $\gamma(D_i)$ represent the time-invariant discontinuity at the cutoff which could be due to time-invariant geographic or administrative sorting, and/or the effects of other policies that change at the border.
- ϵ_{it} represents the potentially unobserved individual-specific characteristics that affect prices.
- $\tau(D_i)$ is the treatment effect of moving from a low-tax to a high-tax HA at a distance D_i to the border.

Empirical strategy: Identifying assumptions

- To identify the treatment effect we need the traditional RD assumptions hold in both periods.:
  1. The functions $f_0(D)$, $f_1(D)$, and $\tau(D)$ are continuous at $D = 0$.
  2. $\mathbb{E}(\epsilon_{it}|D_i = D)$ is continuous at cutoff $D = 0$, for $t = 0, 1$.
- Under thesse assumptions, the treatment effect is identified:

$$\tau(0) = \left(\lim_{D \to 0^+} \mathbb{E}[y_1|D_i = D] - \lim_{D \to 0^-} \mathbb{E}[y_1|D_i = D] \right) - \left(\lim_{D \to 0^+} \mathbb{E}[y_0|D_i = D] - \lim_{D \to 0^-} \mathbb{E}[y_0|D_i = D] \right)$$

- That is, the treatment effect can be interpreted as the difference between the estimated effect of the RD in the second and first periods.

Empirical strategy: Graphical evidence

- We first provide graphical evidence of discontinuities of outcomes around borders.

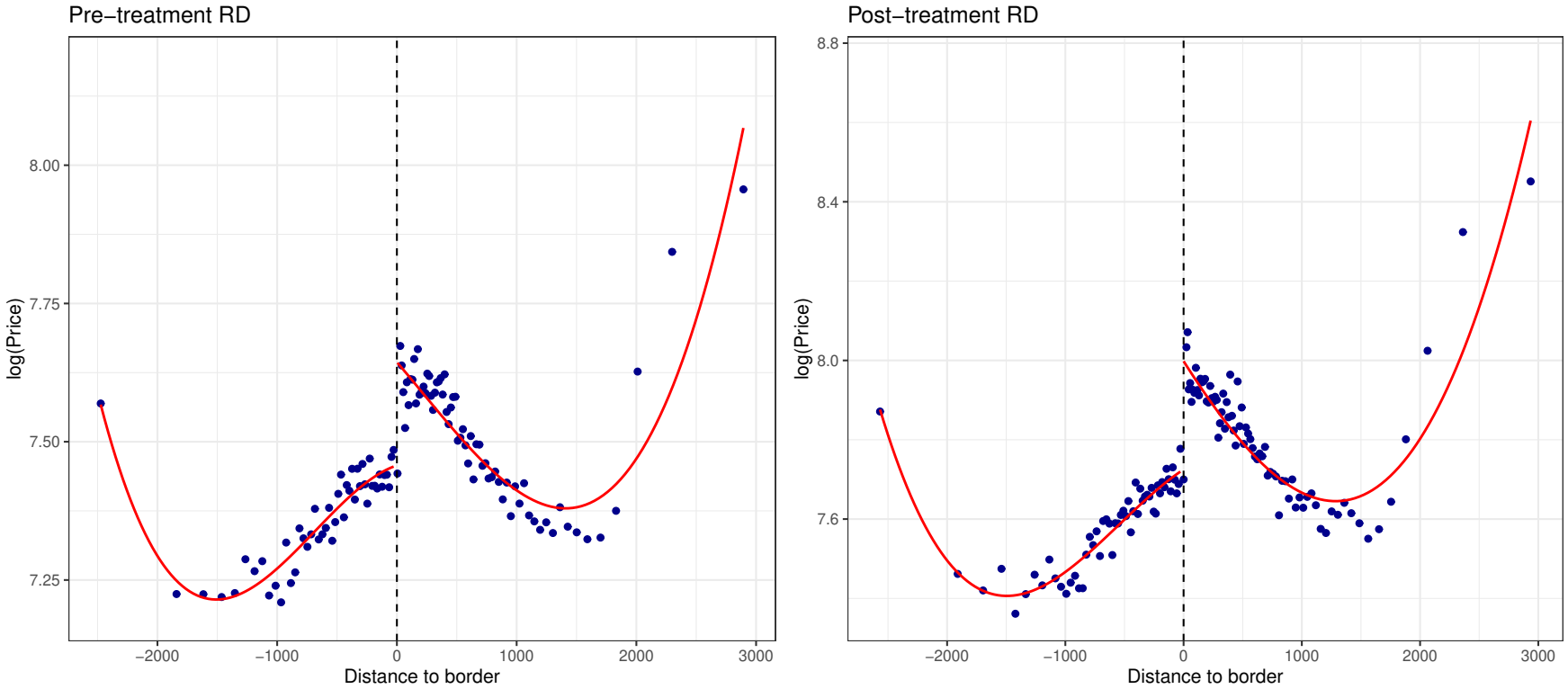

Figure 7: Pre-treatment ($t = 0$) and post-treatment ($t = 1$) RD binned scatterplot

Empirical strategy: Graphical evidence

- The previous figure suggests the existence of **strong administrative sorting**, as we observe a remarkable positive effect in moving from a low-tax to a high-tax HA **before the implementation of HAs**.
- This makes sense, given how HA were created in the first place.

Boletín SII

Noticias SII

Home | Índice de Noticias

Ante el CDE, municipalidades y Subsecretaría de Desarrollo Regional Santiago, 16 de Agosto de 2012

SII detalló metodología técnica utilizada para reavalúo 2013

El Servicio de Impuestos Internos (SII) efectuó el martes 14 de agosto, una presentación acerca del proceso de reavalúo de las propiedades no agrícolas que entrará en vigencia el próximo 1 de enero del año 2013, ante representantes del Consejo de Defensa del Estado (CDE), la Asociación de Municipalidades y la Subsecretaría de Desarrollo Regional.

La reunión detalló la metodología técnica que se está aplicando para definir los valores de los predios a aplicar en el próximo reavalúo, basado en un nuevo estudio urbano que permitió zonificar el territorio en Áreas Homogéneas (AH). Ello corresponde a sectores de las ciudades con características similares en cuanto a uso, categoría constructiva, nivel de densificación y normativa urbana, para los cuales es factible establecer un único valor de suelo, establecido en base al método comparativo de mercado.

Respecto de las construcciones, se expuso la clasificación que efectúa el SII para su incorporación en el catastro, y las modificaciones que se están realizando en los instrumentos que permiten definir su calidad. La valoración de las construcciones se hace sobre los costos de reposición, montos sobre los cuales se aplican los ajustes establecidos en el número 2° del Artículo 4° de la Ley N°17.235.

Asistieron a esta presentación el Presidente del Consejo de Defensa del Estado, Sergio Urrejola, el secretario ejecutivo de la Asociación Chilena de Municipalidades, Enrique Tapia Padilla, y asesores; y el jefe de la División Municipalidades de la Subsecretaría de Desarrollo Regional, Mauricio Cisternas, junto a asesores.

Por el SII, el director Julio Pereira, el subdirector de Avaluaciones, Ernesto Terán, y asesores.

Está contemplado que el SII se reúna próximamente con otros actores relevantes en el proceso, incluyendo a las municipalidades de mayor contribución al Fondo Común Municipal. Se destaca que el Servicio ya se ha reunido con muchas de éstas, explicando la metodología e intercambiando información relevante.

El reavalúo 2013 comprenderá la totalidad de las propiedades no agrícolas del país, alrededor de 5.700.000 bienes raíces.

Empirical strategy: Sorting out administrative sorting

- The previous slide suggests that the government designed HAs in a way that reflects differentials in home value.
- The IRS identifies high-value homes in an area, and then assigns them to high-unit-value HAs. This yields a significant pre-treatment effect. (This can also be interpreted as some form of reverse causality)
- Thus, when we move across the border cutoff, not only is the tax increasing, but also the quality of homes, and this is reflected in the price increase.
- Our objective is to uniquely identify the effect of the tax hike, and so we need to account for the differences in quality of the houses.
- One solution to some of these issues is to use a Differences-in-Discontinuities design (Diff-in-Disc), which removes all time-invariant sorting.
- Additionally, we need to control for the "quality" of the home. We can do this by using the quantile of the home's fiscal valuation.

Empirical strategy: Graphical evidence

- Accounting for this difference yields the following plots:

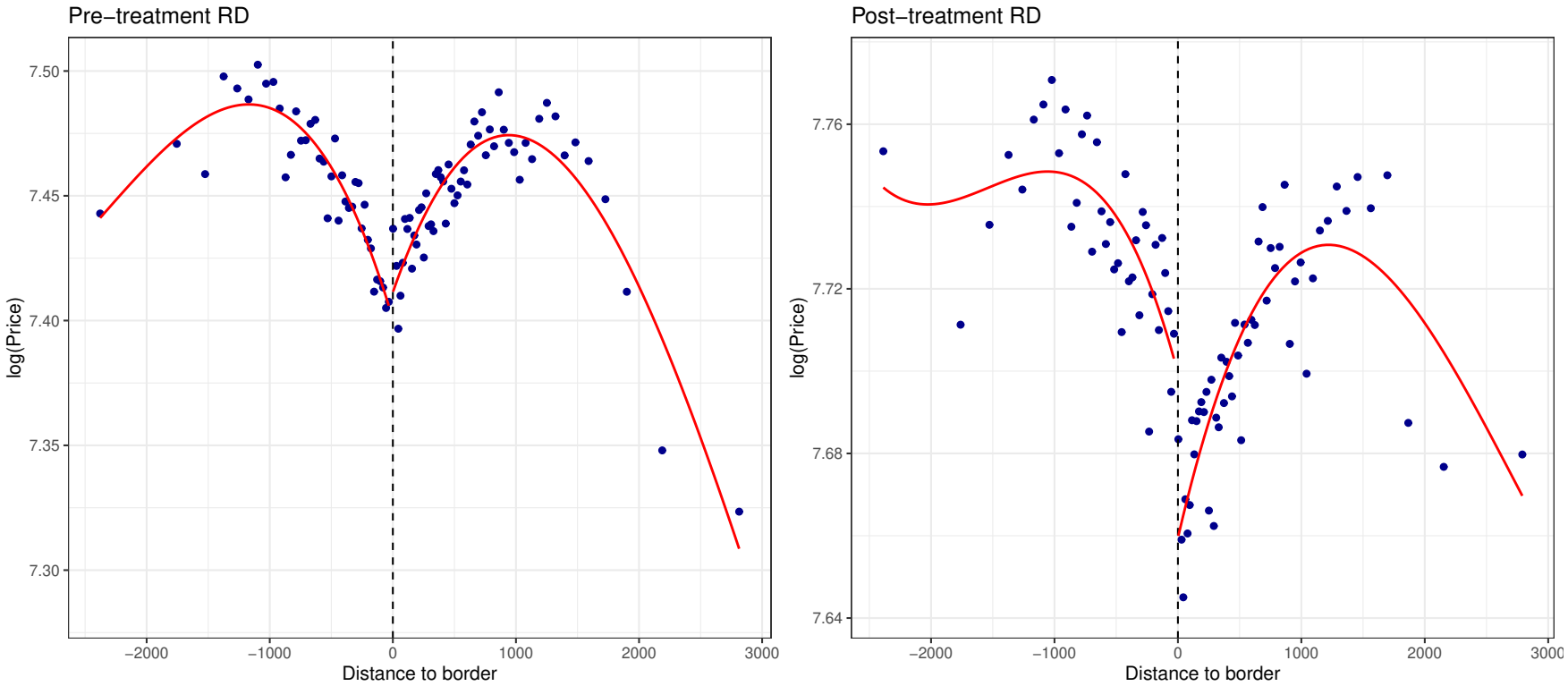

Figure 8: Pre- and post-treatment RD binned scatterplot with fiscal valuation as a control

Empirical strategy: Estimation through Diff-in-Disc

- We estimate the treatment effect in two ways. First, we estimate the following equation through OLS separately for the pre-treament period 2009-2012 ($t = 0$) and for the post-treatment period 2013-2017 ($t = 1$):

$$y_{it} = \alpha_t + \tau_t \mathbb{1}(D_i \geq 0) + \sum_{j=1}^{k} \beta_{jt} D_i^j + \mathbb{1}(D_i \geq 0) \sum_{j=1}^{k} \phi_{jt} D_i^j + \epsilon_{it}$$

  we denote $\hat{\tau}_0$ and $\hat{\tau}_1$ as the estimated effects from periods 0 and 1, respectively. Then, the Diff-in-Disc treatment effect is given by $\hat{\tau} = \hat{\tau}_1 - \hat{\tau}_0$.
- The advantage of this approach is that it allows for traditional optimal bandwidth selection as proposed in Calonico et al. (2014).
- Its disadvantage is that the calculation of standard errors is not straight forward. We decide to use bootstrapped standard errors when employing this method.

Empirical strategy: Estimation through pooled OLS

- Alternatively, we estimate a single model that summarizes four regressions functions of y_{it} on D_{it} (Grembi et al., 2016): two on each side of the threshold, both before and after the intervention. This approach is given by the following equation, which is estimated on a neighborhood of the cutoff point $[0-h,\ 0+h]$:

$$y_{it} = \sum_{j=0}^{k} \delta_{jt} D_i^j + \mathbb{1}(D_i \geq 0) \sum_{j=0}^{k} \gamma_{jt} D_i^j + \mathbb{1}(t=1) \left[\sum_{j=0}^{k} \alpha_{jt} D_i^j + \mathbb{1}(D_i \geq 0) \sum_{j=0}^{k} \beta_{jt} D_i^j \right] + \epsilon_{it}$$

  where $\hat{\beta}_0$ gives the Diff-in-Disc treatment effect.
- The advantage of this approach is that inference is straightforward.
- However, the selection of the bandwidth h is not straightforward.
- Finally, our baseline estimates rely on polynomials of order 1, in line with Gelman and Imbens (2019).

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Baseline results

	Post-treatment	Pre-treament	Diff-in-Disc	Pooled OLS
Estimate	-0.06***	0.0015	-0.0614***	-0.035***
Std. err.	(0.007)	(0.007)	(0.013)	(0.011)
N	92,825	65,055	157,880	129,562

*p<0.1; **p<0.05; ***p<0.01

Table 1: Baseline results

- The post-treatment effect is negative and statistically significant, while the pre-treatment effect is positive and statistically insignificant, suggesting an attenuated administrative sorting effect after controlling for homes' fiscal assessment.
- Both strategies suggest a statistically significant negative effect: moving from a low-tax HA to high-tax HA yields a decrease in price that ranges from 3.5% to 6.1%.

Heterogeneities by price

- We further analyze heterogeneities by home values.
- For example, Elinder and Persson (2017) report that only homes facing high levels of taxation exhibit price drops.

Decile	1	2	3	4	5
Estimate	0.0076	-0.0089**	-0.0177***	-0.0117*	-0.0252**
Std. err.	(0.0089)	(0.0044)	(0.0057)	(0.0062)	(0.0114)
N	25,072	82,397	63,248	72,517	49,480

*p<0.1; **p<0.05; ***p<0.01

Table 2: Heterogeneous effect by price decile, pooled OLS

- Our results suggest that the price effect of border differentials in taxes increases with the value of the home.
- Concretely, homes belonging to the bottom quintile of the price distribution exhibit an effect statistically indistinguishable from zero.
- Other quintiles exhibit statistically significant negative effects, with homes in the top quintile showing the largest price drops.

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

Conclusions and to-do list

- Our results suggest a negative effect of moving from a low-tax to a high-tax HA on prices. The effect ranges from 3.5% to 6.1%.
- We report that high-value homes face stronger negative effects on prices.
- To-do:
  - Expand the discussion on wealth taxation, the incidence of the property tax and its distributional consequences.
  - To do this, we need to compute the average tax hike from moving to a low-tax area to a high-tax area.
- Any further suggestions?

Table of Contents

1. An overview
2. Motivation, related literature and contribution
3. Institutional setting
4. Data sources, assembly and descriptive statistics
5. Empirical strategy
6. Results
7. Conclusions and to-do list
8. References

References I

Bai, C.-E., Li, Q., and Ouyang, M. (2014). Property taxes and home prices: A tale of two cities. *Journal of Econometrics*, 180(1):1–15.

Butts, K. (2021). Geographic difference-in-discontinuities. *Applied Economics Letters*, pages 1–5.

Calonico, S., Cattaneo, M., and Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82:2295–2326.

Correa, J. (2018). Areas homogéneas sii - area metropolitana de santiago 2015.

Elinder, M. and Persson, L. (2017). House price responses to a national property tax reform. *Journal of Economic Behavior Organization*, 144(C):18–39.

Gelman, A. and Imbens, G. (2019). Why high-order polynomials should not be used in regression discontinuity designs. *Journal of Business & Economic Statistics*, 37(3):447–456.

References II

Grembi, V., Nannicini, T., and Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*, 8(3):1–30.

Hilber, C. (2017). The economic implications of house price capitalization: A synthesis. *Real Estate Economics*, 45(2):301–339.

Oates, W. (1969). The effects of property taxes and local public spending on property values: An empirical study of tax capitalization and the tiebout hypothesis. *Journal of Political Economy*, 77(6):957–71.

Oliviero, T. and Scognamiglio, A. (2019). Property tax and property values: Evidence from the 2012 italian tax reform. *European Economic Review*, 118(C):227–251.

Poterba, J. M. (1984). Tax Subsidies to Owner-Occupied Housing: An Asset-Market Approach*. *The Quarterly Journal of Economics*, 99(4):729–752.

References III

Rosen, H. and Fullerton, D. J. (1977). A note on local tax rates, public benefit levels, and property values. *Journal of Political Economy*, 85(2):433–40.